Quorvia MQ 4.2

Administrator Guide

Document QMQ-ADM-4.2, revision C
Applies to Quorvia MQ 4.2.0 through 4.2.3
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Sample document for AI knowledge-retrieval testing. The product, company and all specifications are fictional.

1. About This Guide

This guide is for administrators who install, configure, secure and operate Quorvia MQ, a clustered message broker for AMQP 1.0 and the Quorvia native protocol (QMQP). It covers version 4.2. Commands are shown for Linux; the command-line tool **qmqctl** is installed with the broker.

Conventions: settings appear in `monospace`; values in angle brackets such as <queue> are placeholders. Sizes use binary units (1 MiB = 1,048,576 bytes).

2. System Requirements

Component	Minimum	Recommended
CPU	4 vCPU	8 vCPU
Memory	8 GB RAM	32 GB RAM
Disk	100 GB SSD	500 GB NVMe SSD
Operating system	RHEL 9.2+, Ubuntu 22.04 or 24.04, Windows Server 2022	RHEL 9.4 or Ubuntu 24.04
Runtime	None (native binary, no Java required)	-

Important: Windows Server 2022 is supported for **single-node** installations only. Clusters require Linux on every node. Ubuntu 20.04 is no longer supported as of 4.2.

3. Installation

Quorvia MQ ships as RPM and DEB packages. The service is named **quorvia-mq**. Install, enable and verify the broker:

```
sudo dnf install ./quorvia-mq-4.2.3-1.x86_64.rpm      # RHEL
sudo apt install ./quorvia-mq_4.2.3-1_amd64.deb       # Ubuntu
sudo systemctl enable --now quorvia-mq
qmqctl status
```

Item	Location
Main configuration file	/etc/quorvia/broker.toml
Message data directory	/var/lib/quorvia/data
Snapshots	/var/lib/quorvia/snapshots
Broker log	/var/log/quorvia/broker.log
Initial admin password	/var/lib/quorvia/initial-admin.txt

On first start the broker generates a random password for the built-in **admin** user and writes it to `initial-admin.txt`. The file is deleted automatically after the first successful admin login or after 24 hours, whichever comes first.

4. Network Ports

Port	Protocol	Purpose	Enabled by default
5682	AMQP 1.0 over TLS	Client messaging	Yes
5683	AMQP 1.0 plaintext	Client messaging without TLS	No
7447	QMQP (native)	Quorvia client SDKs	Yes
15690	HTTPS	Admin console and REST API	Yes
9417	HTTP	Prometheus metrics at /metrics	Yes
4369	TCP	Cluster membership (gossip)	Cluster only
25672	TCP	Cluster replication	Cluster only

Note: MQTT is not supported in Quorvia MQ 4.2. MQTT 5 support is planned for version 5.0. The metrics endpoint moved from port 9091 to port 9417 in version 4.2.

5. Configuration Reference

All broker settings live in `/etc/quorvia/broker.toml`. Restart the service after editing. The most frequently changed settings are:

Setting	Default	Notes
max_message_bytes	8388608 (8 MiB)	Hard maximum 134217728 (128 MiB). Larger messages are rejected with QMQ-1004.
retention_hours	72	Range 1-8760. The default changed from 168 to 72 in 4.2.
max_delivery_attempts	5	After the 5th failed delivery the message moves to the dead-letter queue.
consumer_ack_timeout_s	30	Unacknowledged messages are redelivered after this.
max_connections	10000	Per node.
disk_free_alarm_pct	15	Publishers are blocked while free disk is below 15%.
memory_high_watermark	0.6	Fraction of RAM before flow control starts.
tls_min_version	1.3	TLS 1.2 support was removed in 4.1.
snapshot_retain	14	Number of snapshots kept before the oldest is deleted.
api_token_ttl_days	90	Lifetime of REST API tokens.

```
# /etc/quorvia/broker.toml (excerpt)
[limits]
max_message_bytes = 8388608
retention_hours   = 72

[delivery]
max_delivery_attempts  = 5
consumer_ack_timeout_s = 30
```

6. Queues, Dead-Letter Queues and Priorities

Queue names may be up to 200 characters and may contain lowercase letters, digits, dots, underscores and hyphens. Each queue automatically gets a dead-letter queue named **<queue>.dlq**; for example, failed messages from *orders* go to *orders.dlq*.

Messages support priorities 0 to 9, where 9 is the highest. Messages published without a priority get priority 4. Delayed delivery is supported for up to 7 days.

```
qmqctl queue create orders --durable
qmqctl queue stats orders
qmqctl queue purge orders --dry-run      # new in 4.2: shows what would be deleted
```

7. Clustering

Run an odd number of nodes (3 or 5) so that quorum queues keep a majority when a node fails. The **Community edition is limited to 3 nodes**; adding a fourth node fails with QMQ-3100. The Enterprise edition has no node limit and is tested up to 15 nodes.

```
qmqctl cluster join --seed node1.example.internal:4369
qmqctl cluster health
```

Rolling upgrades

1. Upgrade one node at a time.
2. After each node restarts, run `qmqctl cluster health` and wait until it reports **GREEN**.
3. Only then continue with the next node. Never upgrade two nodes at once.

8. Backup and Restore

Snapshots capture queue definitions, users, policies and persisted messages. They do **not** include in-flight messages that are delivered but not yet acknowledged.

```
qmqctl snapshot create --name nightly
qmqctl snapshot list
sudo systemctl stop quorvia-mq
qmqctl snapshot restore --name nightly --force
```

The broker must be stopped before a restore. Snapshots are stored in `/var/lib/quorvia/snapshots` and the newest 14 are kept by default. A snapshot can only be restored to a broker of the same minor version (for example 4.2.x to 4.2.y); otherwise the restore fails with QMQ-2010.

9. Security

- The admin console locks an account for **15 minutes after 5 failed sign-in attempts**.
- REST API tokens expire after 90 days by default (api_token_ttl_days).
- Built-in roles: **viewer** (read-only), **operator** (manage queues and messages) and **admin** (everything, including users and cluster settings).
- The audit log is retained for 400 days.
- Only TLS 1.3 is accepted on TLS listeners.

10. Monitoring

Prometheus metrics are exposed on port 9417 at /metrics. Key metrics:

Metric	Meaning
qmq_queue_depth	Ready messages per queue
qmq_dlq_messages_total	Messages in dead-letter queues
qmq_publish_blocked	1 while publishers are blocked by a disk or memory alarm

Recommended alert: more than 100 messages in any dead-letter queue for 10 minutes.

11. Troubleshooting and Error Codes

Code	Meaning	What to do
QMQ-1001	Authentication failed	Check user name, password or API token.
QMQ-1004	Message larger than max_message_bytes	Reduce the payload or raise max_message_bytes (up to 128 MiB).
QMQ-1007	Publisher blocked by disk alarm	Free disk space or lower disk_free_alarm_pct.
QMQ-1012	Queue not found	Create the queue or fix the queue name.
QMQ-2003	Cluster quorum lost	Bring nodes back until a majority is online. Use qmqctl cluster force-boot only as a last resort: it can lose messages.
QMQ-2010	Snapshot version mismatch	Restore only to the same minor version.
QMQ-3100	License node limit exceeded	Community edition allows 3 nodes; upgrade to Enterprise.

12. Upgrading and Release Notes

> **Important:** Direct upgrades from 3.x to 4.2 are **not supported**. Upgrade to 4.0 first with `qmqctl migrate --from 3.9 --to 4.0`, then upgrade to 4.2.

What changed in 4.2

- Default retention_hours changed from 168 to 72.
- New command: qmqctl queue purge --dry-run.
- Prometheus metrics port moved from 9091 to 9417.
- Ubuntu 20.04 support removed.

4.2.3 patch release

Fixes a memory leak in priority queues (bug QMQ-48211). No configuration changes.

13. Support

Edition	Channel	Severity 1	Severity 2	Severity 3
Enterprise	Phone and portal, 24x7	1 hour	4 business hours	2 business days
Community	Community forum only	No guaranteed response	-	-